

Applied A.I. Solutions

Agile Project Management Methodologies

Quotations, Reference and Recommended Bibliography:

- 1 - "The Epic Guide to Agile – More Business Value On A Predictable Schedule With Scrum" – Author: Dave Todaro - Copyright © 2019 David M. Todaro – Published by R9 Publishing LLC
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Agile Project Management Methodologies

Ritual + Rhythm + Communication + Value +
Commitment + Teamwork + Discipline + Transparency
+ Continuous Improvement + Rapid Development

Week 1 / Day 2

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Scrum

(cont'd)

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Three Artifacts

1. Product Backlog

- Is a simple shared source of truth as it pertains to the vision for the product.
- Represents the entire scope of the product to be produced by the team, typically broken up into releases or versions.
- Included on the product backlog is all the work the Scrum team needs to do:
 - User Stories – represent new functionality to be built
 - Bugs – discovered in user stories that were previously thought to be done
 - Tasks (or chores) – provide the value to the development team

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Three Artifacts – cont'd

2. Sprint Backlog

- Is a set of product backlog items the development team has committed to driving to done during the current sprint, as well as a plan to get the work done in the form of subtasks for each product development item.

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Three Artifacts – cont'd

2. Sprint Backlog includes the following:

- The product backlog items the development team believes they can complete during the sprint
- For each product backlog item, a list of subtasks required to get it to done. Ideally, each subtask could be completed in one day or less.
- For each subtask, an hour estimate of how long the development team thinks it will take to complete it.

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Three Artifacts – cont'd

2. Sprint Backlog – Jira Software:

- Visualizing the work of the sprint with sprint task board.
- Keeping track of who's working on each subtask.
- Understanding how much time is left for each subtask.
- Tracking sprint progress with a sprint burndown chart.

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Three Artifacts – cont'd

2. Sprint Backlog – Jira Software: Task Board

Scrum 1

Scrum Sprint 1

Implement the new weather alert system 🌤️🚨 – and make over 50,000+ customers very happy 😊

🕒 35 days remaining Complete sprint ⋮

🔍 Quick filters ▾ Assignee ▾

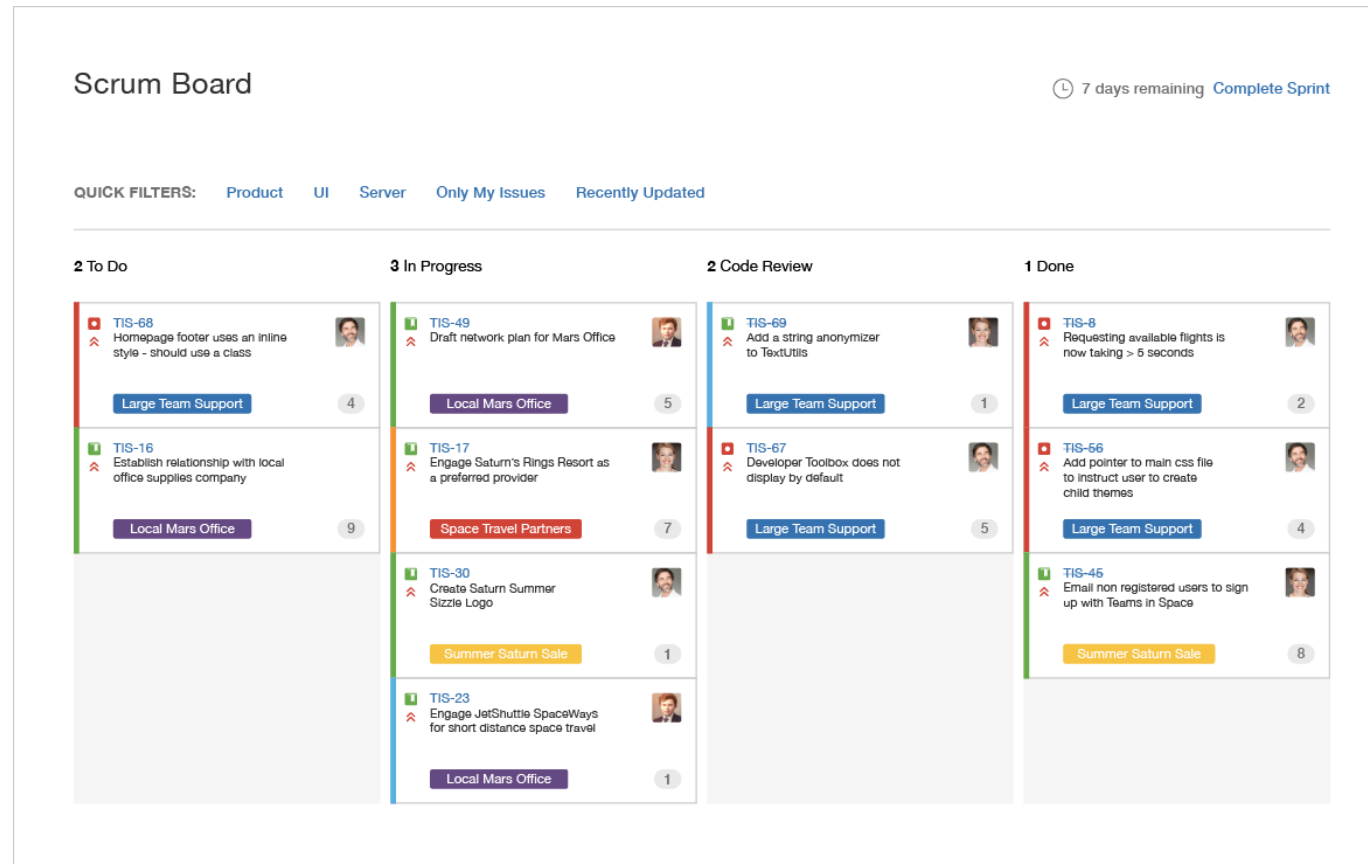
TO DO	IN PROGRESS	DONE
SMART-43 Recalibrate the semi-coherent anomaly i... Make rocket go now 🔧 ⬆️ - SMART-45 Reticulate splines 🔧 ⬆️ - SMART-46 Align the dish 🔧 ⬆️ - SMART-47 SMART-17 Add app alert for changed weather event... MK-33 replicators are down. 🔧 ⬆️ - SMART-40 Adjust API for alert popup 🔧 ⬆️ - SMART-39 Update notifications settings with weather	Add app alert for changed weather events 111 🔧 ⬆️ 4w SMART-17 SMART-3 Push notifications documentation updates Update documentation and push through channels 🔧 ⬆️ - SMART-29 SMART-6 Low-power indicator optimasation on mod... Draw up new schematics for power indicator panel 🔧 ⬆️ 3d SMART-19 Reset power-indicator threshold after fatal shutdown due to low power 🔧 ⬆️ 3h SMART-33	Recalibrate the semi-coherent anomaly in preparation to fluctuate our tachyon catalyst 🔧 ⬆️ - SMART-43 Hello world! 🔧 ⬆️ - SMART-44 SMART-17 Add app alert for changed weather event... Update documentation 🔧 ⬆️ - SMART-28 SMART-3 Push notifications documentation updates Review new copy 🔧 ⬆️ - SMART-35 Low-power indicator optimasation on model B6.2 Epic 123

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Three Artifacts – cont’d

2. Sprint Backlog – Jira Software: Task Board - Subtask Management



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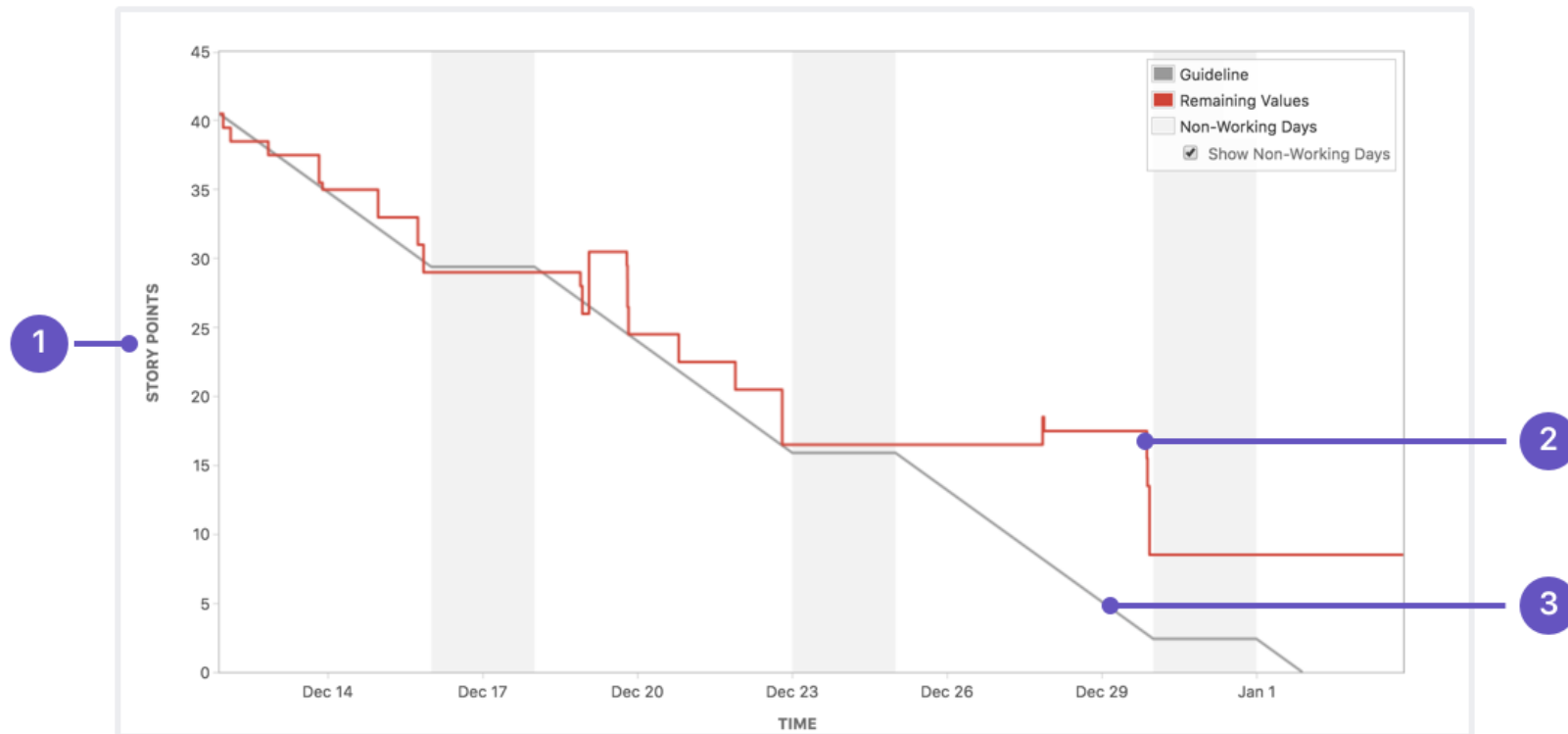
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Three Artifacts – cont'd

2. Sprint Backlog – Jira Software: Burndown Report

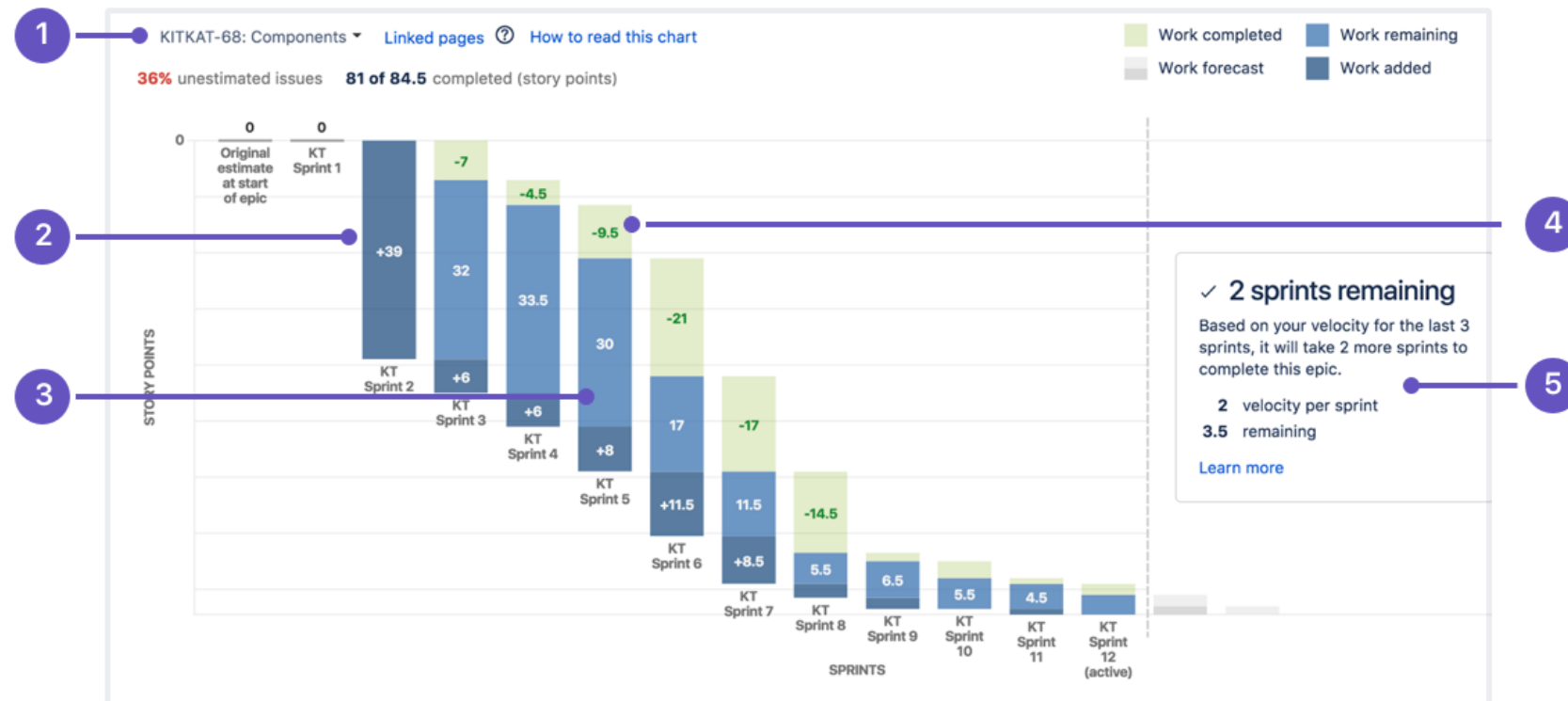


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Three Artifacts – cont'd

3. Product

- It is the end result and is used to describe the output of the team, regardless the of the type of work being done.
- A definition of done is critical tool to ensure everyone knows what “done” means. Its main areas of focus are on quality and completeness.
- The product must be at a shippable level of quality at the conclusion of each sprint.
- It represents a shippable product increment as each sprint builds onto the state of the product at the end of the prior sprint, adding the product backlog items completed during the current sprint.

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Three Artifacts – cont'd

3. Product - Summary

- The product is the collection of all the work to date by the Scrum team from all prior sprints.
- The focus is to build 100% of a small portion of functionality and driving it to a usable state at the end of each sprint.
- “Done” means the product is shippable (with quality and complete), avoiding last minute delays by ensuring everything required has been completed.

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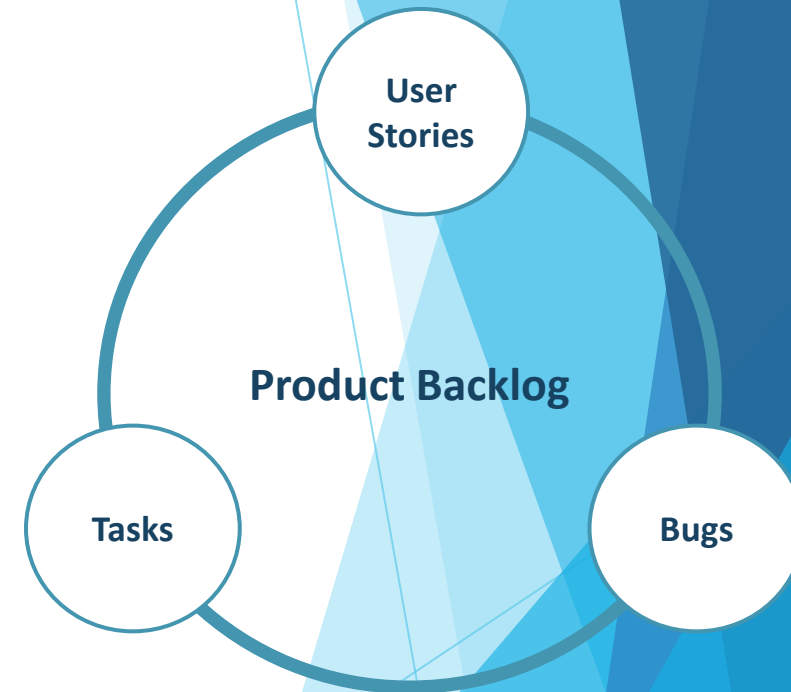
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Product Backlog

1. Single Shared Source of Truth

- **Product Backlog** provides a shared “**Source of Truth**” as it pertains to the vision for the product.
- Allows both the business stakeholders and development team to always understand the vision for the product.
- It is the whole Product wish list, **owned** by the business, and ordered by business value.
- It contains everything the team needs to do to complete the product aligned to the vision of the Product Owner.



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Product Backlog – cont'd

- **User Stories** are lightweight requirements that represent new functionality that delivers value to business stakeholders; including end users.
- **Tasks** represent items the development team would like to do, that provide direct value to the team and indirect value to business stakeholders.
- **Bugs** are defects that made it into the product when it was deemed shippable in a prior sprint, which means bugs that made it through the development team's testing process undetected.

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Product Backlog – cont'd

Order of the Product Backlog

- **When a new product backlog is first created, it will likely only contain user stories.**

For example, for an e-commerce web application, some User Stories can be:

- As a Shopper, I want to check out, so I can get my products shipped to me
- As a Shopper, I want to review my cart, so I can make adjustments prior to checkout
- As a Shopper, I want to view a list of products, so I can select some to purchase
- As a Shopper, I want to login, so I can manage my account
- As an Administrator, I want to modify the list of products, so I can adjust our offerings
- As a Fulfillment Specialist, I want to print a picking report, so I can prepare products to ship
- As a Fulfillment Specialist, I want to print packing labels, so I can ship packages
- As a Finance staff, I want to view analytics about revenue, so I can see how we measure against the goals

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Product Backlog – cont'd

Tasks and Bugs on the Product Backlog

- Once the scrum team is up and running, the development team will ask for some tasks to be added.
- By managing this kind of work on the product backlog, it provides visibility to the product owner and allows him/her to properly prioritize it.
- Once the first shippable product increment is produced, there will inevitably be at least a bug discovered during each sprint.

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2. Product Backlog belongs to the Product Owner.

- The Product Owner is the business representative, embedding the collective intelligence and vision of all stakeholders directly into the team.
- Responsible for understanding the business drivers related to the product.
- Because the Product Owner is the only one who has the perspective of both the business as well the ongoing development of the product, he/she is the only one permitted to decide what is on the product backlog, and in what priority order.

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3. Requirements evolve over time.

- The product backlog is in constant state of evolution.
- The only requirement to lock down a portion of the product backlog is for the scope of work in the very next sprint.

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4. Discussing the backlog with the team is important.

- The process of refining the backlog happens throughout the development of the product.
- Most Scrum teams get together one or two times during each sprint to talk about what's coming up next on the product backlog.
- The Product Owner can explain the vision for each upcoming story providing background, context and how it provides value.
- The Team can ask questions to get more details, increasing their understanding of what the product owner's vision is.

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5. Tracking the projected schedule with a Release Burndown Chart.

- The focus of most Scrum teams is on more a short-term goal, often measured in terms of months.
- The short-term goal is defined by the Product Owner and is aligned with the needs of stakeholders and the organization.
- The goal is represented for a **subset of the product backlog**, starting at the top and including everything the product owner believes should be the next milestone for development of the product.
- This group of product items is called a **Release** and allows tracking progress towards the goal and forecasting a timeframe to reach it.

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5. Tracking the projected schedule with a Release Burndown Chart – cont'd

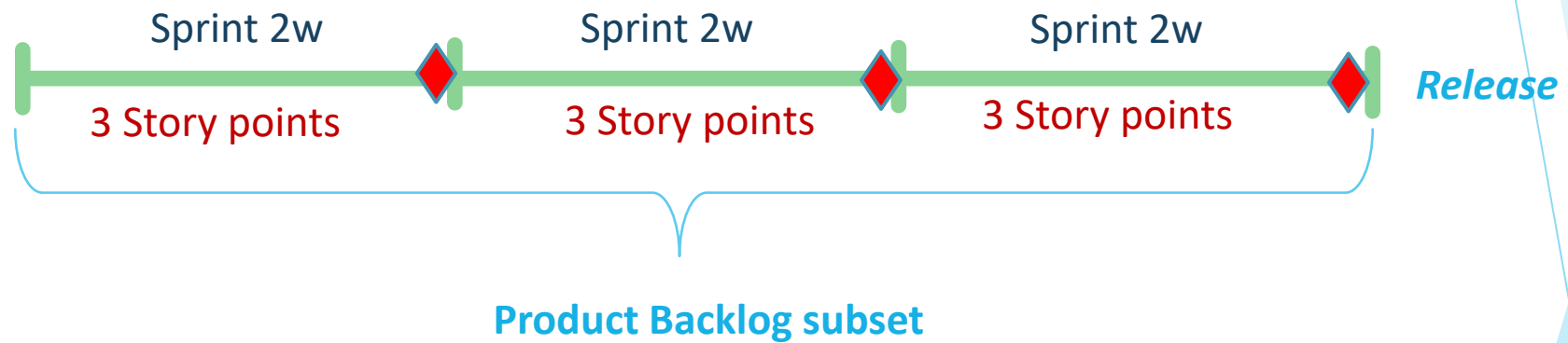
- The amount of work remaining is driven by s size estimate, measure in **Story Points**.
- The speed of the Team is determined by how many story points they have completed during prior sprints, which is called **velocity**.
- Typically, an average velocity of the last three sprints is used for the release burndown.

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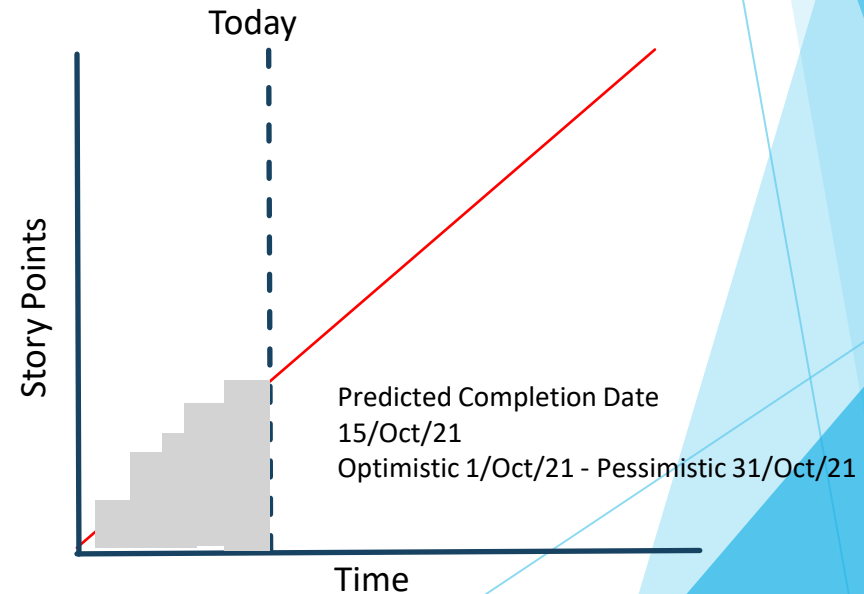
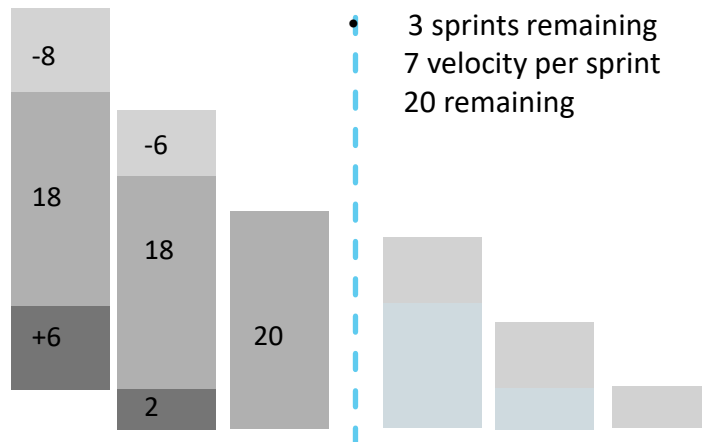


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6. Projecting a Range of Dates with Version Report

- Other types of charts and reports can be used to track progress with the current release.
- Jira** includes different types of reports which uses the same information as the release burndown (velocity of the team and the amount of work remaining).



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User Stories

- Communication is especially important to address with agile projects, with their focus on rapid delivery and adapting to change.
- User Stories provide the basis of communication and are the foundation of shared understanding.
- They facilitate the direct connection between the business and the development team, to ensure that what gets built delivers the highest business value in the shortest time and with shippable quality.

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User Stories – cont'd

1. User Story is a Descriptive Sentence.

Example:

“As a Shopper, I want to check out, so I can get my products shipped to me.”

“As a [role], I want to [capability], so I [benefit]”

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User Stories – cont'd

2. Details are added through conversations.
 - 5 to 15 bullet points can be added to the story to capture details.
 - Product Owner's checklist: this bullet list represents the list of conditions that need to be met to consider the story done and reach the acceptance criteria.
 - The Product Owner must review and accept the work of the development team.

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User Stories – cont'd

3. What, not how.

- It gives the development team more latitude to come up with solutions
- It helps defer the details
- It keeps the user story lightweight

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User Stories – cont'd

4. Each user Story has a size estimate
 - Size, not time
 - Techniques for specifying size
 - The development team determines the size estimate
5. Additional written details are ok
 - Background
 - Notes
 - Technical notes
 - Estimating assumptions
6. Use of traditional specifications (flowcharts, spreadsheets, standards, etc.)

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Example

Story	“As a Bookkeeper, I want to reconcile bank accounts, so I can make sure the bank balance is the application is accurate.”
Background	Customers need to make sure their banking transactions in the application match what’s in their bank account, so they can ensure the bank balance in the app matches reality
Acceptance Criteria	I can import transactions from my bank I can see matching transactions based on dollar amount I can create a new transaction on the fly if it is not in the system yet I can create a new vendor on the fly if it is not in the system yet I can create rules that will automatically classify recurring transactions
Notes	The customer said she would like to show us an example of how she does this by hand today using Excel
Technical Notes	The programmer has used a software package on the prior project (details to follow)
Estimating Assumptions	Transactions will be imported from a file We’ll be able to find a pre-built component, so we do not have to write this from scratch Up to two import formats. If more formats are required, we should re-estimate

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The Sprint

- Scrum uses sprints to address the unpredictable nature of software development.
- It is a short, fixed-duration production cycle, to timebox the work required to complete a set of shippable functionality.
- Scrum comprises the deadline to an extreme degree, reducing it to a sprint lasting one week to a calendar month.
- Everything required to build a small part of the product is completed during this timeframe, including technical design, coding. Testing, bug fixing, and acceptance by the product owner.

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The Sprint – cont'd

Scrum includes the following rules:

- The development team selects the scope to be completed
- The product must be a shippable state at the end of the sprint
- The development team decides how to complete the work
- No scope changes by the Product owner are permitted during the sprint
- The end date of the sprint cannot be changed – positive stress

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The Sprint – cont'd

Scrum includes the following rules:

- The schedule can be forecast to within plus or minus 10% once about four sprints have been completed.
- Product backlog items that are not in the current sprint can be modified at any time by the Product Owner, including changing the order of user stories.
- Everything is done during the sprint
 - Respond to changing conditions
 - Right-size the time investment
 - Discovery of technical challenges
 - Predicting how long it will take

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Scrum process overview

Stakeholders

Vision:
Anticipated
ROI, Releases,
Milestones

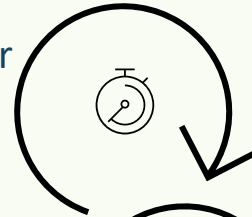


Product Backlog:
emerging, prioritized
requirements

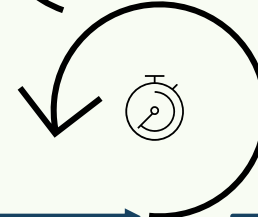
Selected
Product
backlog

Sprint
Backlog

Every 24-hour
inspection

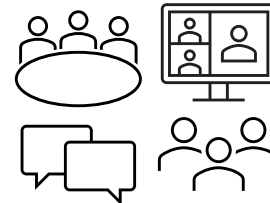


Sprint
iteration



Scrum skeleton

Increment of functionality
New functionality is demonstrated
at the end of Sprint



Scrum team

Quotations, Reference and Recommended Bibliography:

1 - "The Epic Guide to Agile – More Business Value On A Predictable Schedule With Scrum" – Author: Dave Todaro - Copyright © 2019 David M. Todaro – Published by R9 Publishing LLC

2 - "Agile Project Management with Scrum" – Author: Ken Schwaber – Copyright © 2004 by Ken Schwaber – Published by Microsoft Press

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Scrum Ceremonies: Sprint Planning Summary

- Timing: beginning of the sprint
- Attendees: Product Owner, Scrum Master, development team
- Timebox:

Sprints of	1-week		2-week		3-week		4 weeks or calendar month
duration	2 hrs. max		4 hrs. max		6 hrs. max		8 hrs. max

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Scrum Ceremonies: Sprint Planning Summary – cont'd

- Agenda:
 1. The development team determines the **capacity** of the team expressed in hours that they can dedicate to this sprint during its calendar duration.
 2. The Product Owner summarizes the first product backlog item for the development team to refresh their memory.
 3. The development team breaks the product backlog item down into subtasks and drive it to a shippable state (quality + completeness) and estimate the total hours required for each subtask.
 4. The development team reviews the total hours estimated for all the subtasks identified so far and compares it to the total hours available (capacity)

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Scrum Ceremonies: Sprint Planning Summary – cont'd

- Agenda – cont'd:
 5. The development team repeats steps 2-4 until they feel comfortable that the total hours they have estimated can be completed within their capacity.
 6. The development team selects a sprint goal based on the product backlog items that fit into the sprint.
 7. If the project management tool being used requires that the sprint is “started”, the Scrum Master completes the steps to start the sprint.
 8. Each team member volunteers for a subtask or two that they’ll work on between the end of sprint planning and the first daily scrum.

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Scrum Ceremonies: Sprint Planning Summary – cont'd

- Outcomes:
 1. The sprint backlog has been created with the product backlog items broken down into subtasks, each with an hour estimate.
 2. A sprint goal has been determined and written down.
 3. Any steps to start a sprint in the project management tool being used have been completed.
 4. Each development team member has volunteered for the first subtask they'll be focus on completing prior to the next daily scrum.

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Scrum Ceremonies: Daily Scrum Summary

- Timing: every working day, ideally at the same time every day.
- Attendees: Product owner, Scrum Master, Development Team
- Timebox: 15 minutes
- Agenda
 1. Update. Each development team member answers the following: What have you gotten done since the last daily scrum? What are you committing to getting done before the next daily scrum? What impediments are in the way?
 2. Capture of items requiring additional discussion.
 3. Sprint Burndown Chart review.

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Scrum Ceremonies: Daily Scrum Summary – cont'd

- Outcomes
 1. The team members understand the status of the work of the sprint
 2. Every development team member has made a **commitment** to the other team members about how they will do their part to drive the sprint forward, by volunteering for one or more subtasks they will complete before the next daily scrum
 3. The Scrum Master volunteered to remove impediments that were previously identified.

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Scrum Ceremonies: Sprint Review Summary

- Timing: at the end of each sprint.
- Attendees: Product owner, Scrum Master, Development Team, and Key Stakeholders invitee by the Product Owner
- Timebox:
 - 1-week sprints: 1 hour max (recommended)
 - 2-week sprints: 2 hours max (recommended)
 - 3-week sprints: 3 hours max
 - 4 weeks or a calendar month sprints: 4 hours max

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Scrum Ceremonies: Sprint Review Summary – cont'd

- Agenda
 1. Discussion about what product backlog items the development team committed to completing at the start of the sprint, and the sprint goal.
 2. Discussion about whether everything was completed and, if not, why?
 3. Discussion about what went well and what problems were encountered.
 4. A demonstration of the stories that were driven to done by the development team.
 5. A review of what is coming next on the Product Backlog, including feedback from the stakeholders about things missing or factors that may change the priority order.
 6. The projected completion timeline.

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Scrum Ceremonies: Sprint Review Summary – cont'd

- Outcomes:
 1. The stakeholders understand the status of the development of the product.
 2. The Scrum team has heard the feedback from the stakeholders.
 3. Anything that may impact the product backlog has been discussed, especially anything that may affect the product backlog items that may make it into the next sprint.

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Scrum Ceremonies: Retrospective Summary

- Timing: soon after the sprint review, and prior to the next sprint planning ceremonies
- Attendees: Product owner, Scrum Master, Development Team
- Timebox:
 - 1-week sprints: 45 minutes max (recommended)
 - 2-week sprints: 1.5 hours max (recommended)
 - 3-week sprints: 2 – 2.5 hours max
 - 4 weeks or a calendar month sprints: 3 hours max

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Scrum Ceremonies: Retrospective Summary – cont'd

- Agenda
 1. Discussion about how things went during the sprint:
 - What went well?
 - What could have gone better?
 2. Discussion of ideas on how to improve the process, and experiments to try during the upcoming sprint.
- Outcomes
 1. The team has a shared understanding of how the sprint went.
 2. One or more things to try during the upcoming sprint have been identified.

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Scrum Ceremonies: optional, as needed

- Parking Lot
- Backlog Refinement
- Backlog Estimating
- Backlog Prioritization

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Ceremony Schedule (Typical)

Monday	Tuesday	Wednesday	Thursday	Friday
Sprint Planning	Daily Scrum Parking Lot	Daily Scrum Parking Lot	Daily Scrum Parking Lot	Daily Scrum Parking Lot
Daily Scrum Parking Lot Backlog Prioritization	Daily Scrum Parking Lot Backlog Refinement	Daily Scrum Parking Lot Backlog Estimating	Daily Scrum Parking Lot	Daily Scrum Parking Lot Sprint Review Retrospective

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End of Week 1 / Day 2

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